



Course Plan: Project Planning & Prototyping

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247-519-VA

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COURSE PLAN

Course Identification

Course title	Project Planning & Prototyping
Course number	247-519-VA
Credits	3 (3-2-3: 3 hours lecture, 2 hours lab, 3 hours homework)
Prerequisite	247-418-VA (Embedded Systems II) or instructor consent
Term	Fifth semester, Computer Engineering Technology
Instructor	J. F. N. Salik
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Course Description

Students develop and validate a functioning IoT prototype subsystem over one semester, applying industry product-development methods at each stage. The course positions itself within the *Stage-Gate* model used by engineering firms: students enter at Gate 2 (concept selected) and exit at Gate 3 (prototype validated and documented). Every chapter delivers one *Portfolio Entry* — a short, structured document section — that accumulates into a complete prototype-documentation package submitted at Milestone 3.

Three graded milestones track progress:

1. **Milestone 1 — Specification Review (end of Week 6):** Portfolio Entries 1–9 assembled. Design decisions locked.
2. **Milestone 2 — Build Review (end of Week 9):** Prototype built and validated against specifications. Planning artefacts complete.
3. **Milestone 3 — Gate 3 Presentation (end of Week 12):** Final live presentation (10–15 min) plus complete documentation package submitted. This is the Gate 3 exit submission.

Running example. All lecture examples use a single IoT robotic sorting subsystem throughout the course (selling price $p = \$1,200$ CAD, variable cost $c_{\text{var}} = \$480$ CAD,

BOM = \$124.85 CAD, $C_{\text{fixed}} = \$8,000$ CAD). Students apply the same methods to their own chosen project.

Stage-Gate Context

Figure 1 places the course within the five-stage product-development model. Students enter with a defined concept (Gate 2) and spend the semester building the evidence needed to pass Gate 3: a validated prototype, a costed BOM, and a documented design history file.

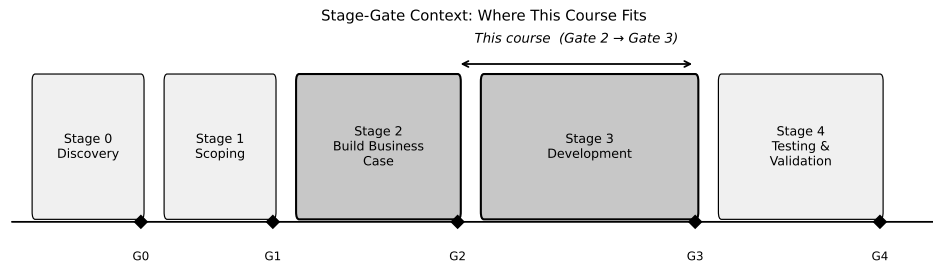


Figure 1: The course spans Stage 2 (Build Business Case) and Stage 3 (Development), exiting at Gate 3 with a validated prototype.

12-Week Timeline

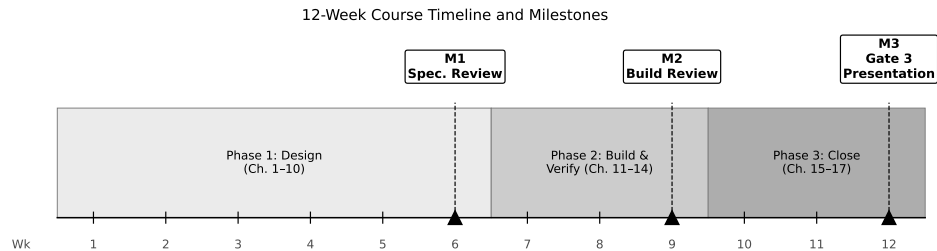


Figure 2: Three course phases and milestone dates. Each phase closes with a graded submission.

Weekly Schedule

The table below lists lecture topics, associated textbook chapters, lab activities, and the Portfolio Entry due at the end of each week. Lab sessions (2 h) are used to work on the current Portfolio Entry under instructor supervision.

Week	Topic	Ch.	Lab Activity	Deliverable
1	Course introduction; product vs. prototype; revenue models; gross margin	1	Project kick-off: choose design project; compute m and GM% for three revenue scenarios	Portfolio Entry 1: Project Context Brief
2	Stage-Gate model; Kano model; voice-of-customer mapping	2	Map five customer needs to Kano categories; identify must-be and performance attributes	Portfolio Entry 2: Stage-Gate + Kano Analysis
3	Seven-stage prototyping workflow; fidelity ladder; effort model $E(f) = E_0 \cdot k^f$	3	Select fidelity level for your project PoC; compute effort ratio $E(5)/E(2)$	Portfolio Entry 3: Workflow Plan
4	Writing engineering specifications; MoSCoW; tolerance vs. requirement	4	Draft five MoSCoW specifications for your project; set measurable pass/fail criteria	Portfolio Entry 4: Specification Table
5	System architecture; block diagrams; interface definition; specify-and-trust vs. prototype	5	Draw block diagram; classify each subsystem using the three-question decision flow	Portfolio Entry 5: System Architecture
6	Component selection; datasheet reading; supplier comparison; risk ranking	6	Build a three-candidate comparison table for one key component; select and justify	Portfolio Entry 6: Component Selection

Milestone 1 — Specification Review (end of Week 6)

Assemble Portfolio Entries 1–6; submit as one PDF. All design decisions locked from this point.

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Week	Topic	Ch.	Lab Activity	Deliverable
7	Design for manufacturing; DFM, DFA, DFT principles; cost-of-change curve	7	Apply DFM checklist to your PCB layout; identify one DFM improvement	Portfolio Entry 7: DFX Review
8	AI integration; edge inference; model selection; latency and accuracy trade-offs	8	Run a pre-trained model on target hardware; measure inference latency	Portfolio Entry 8: AI Integration Notes
9	IoT and robotics integration; communication protocols; power budgeting; safety	9	Build power budget table for your project; identify single-point failure modes	Portfolio Entry 9: IoT/Robotics Integration
Milestone 2 — Build Review (end of Week 9)				
Prototype operational; add Portfolio Entries 7–9 to package. Instructor sign-off required.				
10	Project planning; PERT; Gantt charts; critical path	10	Build a five-task PERT network; compute t_e and σ^2 for each task	Portfolio Entry 10: Project Schedule
11	BOM and procurement; as-built BOM; supplier lead times; RFQ process	11	Update your BOM with actual component costs; flag lead-time risks	Portfolio Entry 11: As-Built BOM
12	Engineering economics; break-even; fixed vs. variable cost; GM% sensitivity	12	Compute N^* and GM% for your project at three production volumes	Portfolio Entry 12: Cost Analysis
13	Version control and design history file; DHF structure; change-control log	13	Set up a Git repository; write one ECO entry for a design change	Portfolio Entry 13: Version Control Setup
14	Validation testing; statistical process control; fault tree analysis	14	Run 10 measurements of a key spec; compute $\bar{x}$, s , z , C_p	Portfolio Entry 14: Validation Report

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Week	Topic	Ch.	Lab Activity	Deliverable
15	Intellectual property; patents; trade secrets; NDAs; prior-art search	15	Complete IP checklist for your project; identify one protectable element	Portfolio Entry 15: IP Checklist
16	Documentation assembly; traceability matrix; requirements coverage; DHF review	16	Complete the 14-section portfolio package; run coverage audit	Portfolio Entry 16: Complete Documentation Package
17	Cost bridge to production; unit-cost hyperbola; revenue break-even N_{be} ; Gate 3 readiness statement	17	Compute C_{unit} at $N = 1, 50, 500$; write Gate 3 readiness statement	Portfolio Entry 17: Cost Projection + Gate 3 Statement

Milestone 3 — Gate 3 Presentation (end of Week 12 exam period)

10–15 min live presentation + complete documentation package. This is the Gate 3 exit submission.

Evaluation Summary

Assessment	Weight	Due
Milestone 1 — Specification Review	20%	End of Week 6
Milestone 2 — Build Review	20%	End of Week 9
Weekly Portfolio Entries (10–16, 17)	25%	Each week (lab submission)
Milestone 3 — Gate 3 Presentation	35%	Exam period
Total	100%	

Portfolio Entry policy. Each Portfolio Entry is submitted at the end of its lab session. Entries submitted late without prior arrangement are capped at 50%. Entries are cumulative: later submissions may revise earlier ones in response to instructor feedback, provided the revised section is clearly marked [Rev. N, date].

Required Materials

- *Project Planning & Prototyping* (this textbook), J. F. N. Salik, 2026. Available on Lea / Moodle as PDF.
- A prototyping kit issued at the start of term (contents listed in Ch. 6 BOM).
- Laptop capable of running Python 3 and KiCad 8.

- GitHub account (free); Git client installed locally.

SPLI Note

This course is structured to support *Self-Paced Learning Integration* (SPLI). Students who complete a chapter's Portfolio Entry to a passing standard before the scheduled week may proceed to the next chapter independently. Milestone dates are fixed for all students regardless of pace. Students working ahead are encouraged to iterate on earlier Portfolio Entries using subsequent chapters' methods.